Hongyu Xiao

Curriculum Vitae

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Education

2019 - Present **B.Sc. in Honours Neuroscience**, McMaster University, Canada, GPA: 11.64/12 3.94/4

Research Interests

- **Brain-Computer Interfaces and neuroengineering**
- **Computational neuroscience**
- **Human tactile and rhythmic perception based on Bayesian inference models**
- **Influence of gut microbiome on the gut-brain axis**

Honors & Awards

2020,2022,2023 **Deans' Honour List**

2021-2023 **Department of Biology Honour Roll**

2019 **The McMaster President's Award Scholarship**

Research Experiences

May 2023 - Aug 2023 **Lab Assistant, Intern**, *Dr. Daniel Taliun*, Taliun's Lab at the Department of Human Genetics, McGill University

- Develop computational algorithms and analytical tools for studying human genetics

Nov 2022 - Present **Lab Assistant**, *Dr. Jonathan Cannon*, TRIMBA Lab at the Department of Psychology, Neuroscience and Behaviour, McMaster University

- Study whether both motor system and intrinsic rhythmic clock contribute to human rhythmic perception
- Built an analysis pipeline for rhythmic data using R
- Building computational models to study the sense of rhythm

Sep 2022 - Apr 2023 **Lab Assistant**, *Dr. Daniel Goldreich*, Department of Psychology, Neuroscience and Behaviour, McMaster University

- Study whether human haptic categorization follows a Bayesian optimal manner

- Conducted human sensory tests, analyzed sensory performance data, coded in the LabVIEW programming environment
- Read and evaluated perceptual neuroscience literature with an emphasis on tactile and computational neuroscience
- Gained an understanding of the Bayesian perceptual framework, signal detection theory, psychophysical testing methodologies, automated experimental design, and data analysis

Aug 2022 **Research Student**, *Dr. Dejan Markovic*, Department of Electrical and Computer Engineering at the University of California, Los Angeles, and Center for NeuroTechnology

- Systemically studied the basic logic behind Brain-Computer Interfaces, including neuroengineering, membrane biophysics, and neuroscience
- Investigated in the insights into human memory and cognition
- Discussed the underlying logic of neural recording and stimulation
- Gave a presentation discussing the response patterns of the cochlear nucleus neurons based on integrate-and-fire model and temporal coding model

Aug 2021 **Research Student**, *Dr. Sreekanth Chalasani*, The Salk Institute for Biological Studies Molecular Neurobiology Laboratory

- Systemically studied neurological diseases, such as schizophrenia and anxiety disorders
- Applied CRISPR technology to try to fix a single-point mutation(position 2207) on the SLC6A4 gene, in order to generate a hypothesis of the effects of mutations on the SLC6A4 gene on anxiety disorder
- As a group leader, lead the group to give a presentation describing the connection between SLC6A4 gene mutation and the generation of anxiety disorder

Sept 2016 **Research Assistant, Intern**, *Dr. Chao Liu*, Bioinformatics Team of Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China.

- Investigated proteomics, which contributed to the analysis of big data of protein. Efficiently assisted in conducting Peptide Mass Fingerprinting to help to analyze protein categories

Published Papers

Xiao, H. (2022, July). Impact of gut microbiome on anxiety. In AIP Conference Proceedings (Vol. 2511, No. 1, p. 020059). AIP Publishing LLC. https://doi.org/10.1063/5.0094814

Extracurricular Activities

2022 - Present **Meducator MeduCollab Team**, *Undergraduate Health Sciences Journal*, McMaster University, ON

- Participate in research and writing workshops

- Writing articles on a Health Sciences topic

2022 - Present **Science Student Ambassador Team**, *The Faculty of Science Recruitment Team*, McMaster University, ON

- Assist with undergraduate recruitment and engagement opportunities both virtually and in-person

2021 - Present **A Med Thing or Two**, *Non-profit organization*, Hamilton, ON

- Project Lead in the Production & Social Media Team
- Contribution to a platform that provided medical and biological information involving simplified mechanisms and recent research on social media (e.g. WeChat official account)
- Job content:
  - Lead any project of interest and its production
  - Provide project outline for Directors of Production and Education for broadcast scheduling
  - Coordinate the research, writing, translation, and final presentation of the leading project translation
  - Lead in the production of weekly content on all social media platforms (TikTok, Instagram, WeChat videos)
  - Provide any additional support essential to the operation of AMToT or his/her/their team

2018- 2019 **The Scarborough-Agincourt Federal Liberal Association**, *Scarborough, ON*

- Contribution to effectively organizing community events (e.g. elections, Canada Day parades, and community parties), with great satisfaction from the visitors

Skills

- **Computational Skills**: Unix/Linux, C/C++, Python, R/Rstudio, Microsoft Excel, LabChart, PyMOL, NetLogo
- **Organizational Skills**: Effective Time Management, Multitask Capacity, Active Listening, Critical Thinking
- **Lab Skills**: Micropipetting, Using Microscope, Spectrophotometer, Biosafety Training, WHMIS Traning, IR Spectrum Analysis, Proton NMR Spectrum Analysis, C-13 NMR Spectrum, Crayfish Dissection, Extracellular and Intracellular Recording, PowerLab Hardware

Languages

- **English:** fluent
- **Chinese:** native level

Interests

Music	Guitar Player of two bands
	Leader and Organizer of a music club in high school
Media Art	Video Editing (Final Cut Pro), Adobe Illustrator, Photography